In [1]:

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

data = {
    "Order_ID": [
        1001, 1002, 1003, 1004, 1005, 1006,
        1007, 1008, 1009, 1010, 1011, 1012,
        1013, 1014, 1015, 1016, 1003, 1018
    ],

    "Customer_Name": [
        "  rahul sharma ", "PRIYA REDDY", " Amit Kumar ",
        "sneha rao", None, "  KIRAN  ",
        "ananya reddy", "RAHUL SHARMA", "  arjun reddy ",
        "meena devi", "  ", "Vikram Rao",
        "sai krishna", "  Priya Reddy ", "naveen kumar",
        "divya sharma", " Amit Kumar ", "Rohit Verma"
    ],

    "Product": [
        " laptop ", "MOBILE", "headphones",
        " TABLET ", "Laptop", " mobile ",
        "Headphones", " laptop", "TABLET",
        " laptop ", None, "MOBILE",
        "headphones", " tablet ", "Laptop",
        "mobile", "headphones", " LAPTOP "
    ],

    "Quantity": [
        2, 3, 5, 1, 4, "five",
        2, 3, 1, 2, 6, 4,
        np.nan, 2, 3, 5, 5, 2
    ],

    "Unit_Price": [
        55000, 18000, 1500, 22000, 55000, 18000,
        1500, "unknown", 22000, 55000, 18000, np.nan,
        1500, 22000, 55000, 1558000, 1500, 99999
    ],

    "Order_Date": [
        "2026-01-05", "06/01/2026", "2026-01-07",
        "08/01/2026", "2026-01-09", "10/01/2026",
        "2026-01-11", "12/01/2026", "31/02/2026",
        "2026-01-14", "15/01/2026", "2026-01-16",
        "17/01/2026", "2026-01-18", "19/01/2026",
        "2026-01-20", "2026-01-07", "not-a-date"
    ],

    "Region": [
        " hyderabad ", "KHAMMAM", "warangal",
        " HYDERABAD", "khammam ", "Warangal",
        "hyderabad", " KHAMMAM ", "warangal",
        "HYDERABAD", None, " khammam",
        "Warangal ", "hyderabad", " KHAMMAM",
        "warangal", "warangal", "HYDERABAD"
    ]
}

df = pd.DataFrame(data)

df
```

Out[1]:

	Order_ID	Customer_Name	Product	Quantity	Unit_Price	Order_Date	Region
0	1001	rahul sharma	laptop	2	55000	2026-01-05	hyderabad
1	1002	PRIYA REDDY	MOBILE	3	18000	06/01/2026	KHAMMAM
2	1003	Amit Kumar	headphones	5	1500	2026-01-07	warangal
3	1004	sneha rao	TABLET	1	22000	08/01/2026	HYDERABAD
4	1005	NaN	Laptop	4	55000	2026-01-09	khammam
5	1006	KIRAN	mobile	five	18000	10/01/2026	Warangal
6	1007	ananya reddy	Headphones	2	1500	2026-01-11	hyderabad
7	1008	RAHUL SHARMA	laptop	3	unknown	12/01/2026	KHAMMAM
8	1009	arjun reddy	TABLET	1	22000	31/02/2026	warangal
9	1010	meena devi	laptop	2	55000	2026-01-14	HYDERABAD
10	1011		NaN	6	18000	15/01/2026	NaN
11	1012	Vikram Rao	MOBILE	4	NaN	2026-01-16	khammam
12	1013	sai krishna	headphones	NaN	1500	17/01/2026	Warangal
13	1014	Priya Reddy	tablet	2	22000	2026-01-18	hyderabad
14	1015	naveen kumar	Laptop	3	55000	19/01/2026	KHAMMAM
15	1016	divya sharma	mobile	5	1558000	2026-01-20	warangal
16	1003	Amit Kumar	headphones	5	1500	2026-01-07	warangal
17	1018	Rohit Verma	LAPTOP	2	99999	not-a-date	HYDERABAD

In [2]:
```
df.head()
```

Out[2]:

	Order_ID	Customer_Name	Product	Quantity	Unit_Price	Order_Date	Region
0	1001	rahul sharma	laptop	2	55000	2026-01-05	hyderabad
1	1002	PRIYA REDDY	MOBILE	3	18000	06/01/2026	KHAMMAM
2	1003	Amit Kumar	headphones	5	1500	2026-01-07	warangal
3	1004	sneha rao	TABLET	1	22000	08/01/2026	HYDERABAD
4	1005	NaN	Laptop	4	55000	2026-01-09	khammam

In [3]:
```
df.shape
```

Out[3]:
```
(18, 7)
```

In [4]:
```
df.columns
```

Out[4]:
```
Index(['Order_ID', 'Customer_Name', 'Product', 'Quantity', 'Unit_Price',
       'Order_Date', 'Region'],
      dtype='str')
```

In [5]:
```
df.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 18 entries, 0 to 17
Data columns (total 7 columns):
 #   Column         Non-Null Count  Dtype
---  ------         --------------  -----
 0   Order_ID       18 non-null     int64
 1   Customer_Name  17 non-null     str
 2   Product        17 non-null     str
 3   Quantity       17 non-null     object
 4   Unit_Price     17 non-null     object
 5   Order_Date     18 non-null     str
 6   Region         17 non-null     str
dtypes: int64(1), object(2), str(4)
memory usage: 1.1+ KB
```

In [6]:
```
df.describe()
```

Out[6]:

	Order_ID
count	18.000000
mean	1008.722222
std	5.199610
min	1001.000000
25%	1004.250000
50%	1008.500000
75%	1012.750000
max	1018.000000

In [7]:
```
df.describe(include='all')
```

Out[7]:

	Order_ID	Customer_Name	Product	Quantity	Unit_Price	Order_Date	Region
count	18.000000	17	17	17.0	17.0	18	17
unique	NaN	16	12	7.0	7.0	17	12
top	NaN	Amit Kumar	headphones	2.0	55000.0	2026-01-07	warangal
freq	NaN	2	3	5.0	4.0	2	4
mean	1008.722222	NaN	NaN	NaN	NaN	NaN	NaN
std	5.199610	NaN	NaN	NaN	NaN	NaN	NaN
min	1001.000000	NaN	NaN	NaN	NaN	NaN	NaN
25%	1004.250000	NaN	NaN	NaN	NaN	NaN	NaN
50%	1008.500000	NaN	NaN	NaN	NaN	NaN	NaN
75%	1012.750000	NaN	NaN	NaN	NaN	NaN	NaN
max	1018.000000	NaN	NaN	NaN	NaN	NaN	NaN

In [8]:
```
df.dtypes
```

Out[8]:
```
Order_ID          int64
Customer_Name       str
Product             str
Quantity         object
Unit_Price       object
Order_Date          str
Region              str
dtype: object
```

In [9]:
```
df.isnull()
```

Out[9]:

	Order_ID	Customer_Name	Product	Quantity	Unit_Price	Order_Date	Region
0	False	False	False	False	False	False	False
1	False	False	False	False	False	False	False
2	False	False	False	False	False	False	False
3	False	False	False	False	False	False	False
4	False	True	False	False	False	False	False
5	False	False	False	False	False	False	False
6	False	False	False	False	False	False	False
7	False	False	False	False	False	False	False
8	False	False	False	False	False	False	False
9	False	False	False	False	False	False	False
10	False	False	True	False	False	False	True
11	False	False	False	False	True	False	False
12	False	False	False	True	False	False	False
13	False	False	False	False	False	False	False
14	False	False	False	False	False	False	False
15	False	False	False	False	False	False	False
16	False	False	False	False	False	False	False
17	False	False	False	False	False	False	False

In [10]:
```
df.isnull().sum()
```

Out[10]:
```
Order_ID         0
Customer_Name    1
Product          1
Quantity         1
Unit_Price       1
Order_Date       0
Region           1
dtype: int64
```

In [11]:
```
df.duplicated()
```

Out[11]:
```
0     False
1     False
2     False
3     False
4     False
5     False
6     False
7     False
8     False
9     False
10    False
11    False
12    False
13    False
14    False
15    False
16     True
17    False
dtype: bool
```

In [12]:
```
df.duplicated().sum()
```

Out[12]:
```
np.int64(1)
```

In [13]:
```
df[df.duplicated]
```

Out[13]:

	Order_ID	Customer_Name	Product	Quantity	Unit_Price	Order_Date	Region
16	1003	Amit Kumar	headphones	5	1500	2026-01-07	warangal

In [14]:
```
df[df.duplicated(keep=False)]
```

Out[14]:

	Order_ID	Customer_Name	Product	Quantity	Unit_Price	Order_Date	Region
2	1003	Amit Kumar	headphones	5	1500	2026-01-07	warangal
16	1003	Amit Kumar	headphones	5	1500	2026-01-07	warangal

In [15]:
```
df = df.drop_duplicates()
print("Shape after removing duplicates:", df.shape)
```

```
Shape after removing duplicates: (17, 7)
```

In [16]:
```
df["Quantity"] = pd.to_numeric(
    df["Quantity"], errors="coerce"
)
df["Unit_Price"] = pd.to_numeric(
    df["Unit_Price"], errors="coerce"
)
df[["Quantity", "Unit_Price"]]
```

Out[16]:

	Quantity	Unit_Price
0	2.0	55000.0
1	3.0	18000.0
2	5.0	1500.0
3	1.0	22000.0
4	4.0	55000.0
5	NaN	18000.0
6	2.0	1500.0
7	3.0	NaN
8	1.0	22000.0
9	2.0	55000.0
10	6.0	18000.0
11	4.0	NaN
12	NaN	1500.0
13	2.0	22000.0
14	3.0	55000.0
15	5.0	1558000.0
17	2.0	99999.0

In [17]:
```
df.dtypes
```

Out[17]:
```
Order_ID           int64
Customer_Name        str
Product              str
Quantity         float64
Unit_Price       float64
Order_Date           str
Region               str
dtype: object
```

In [18]:
```
df["Order_Date"] = pd.to_datetime(
    df["Order_Date"],
    format="mixed",
    dayfirst=True,
    errors="coerce"
)
df[["Order_ID", "Order_Date"]]
```

Out[18]:

	Order_ID	Order_Date
0	1001	2026-05-01
1	1002	2026-01-06
2	1003	2026-07-01
3	1004	2026-01-08
4	1005	2026-09-01
5	1006	2026-01-10
6	1007	2026-11-01
7	1008	2026-01-12
8	1009	NaT
9	1010	2026-01-14
10	1011	2026-01-15
11	1012	2026-01-16
12	1013	2026-01-17
13	1014	2026-01-18
14	1015	2026-01-19
15	1016	2026-01-20
17	1018	NaT

In [19]:
```
df.dtypes
```

Out[19]:
```
Order_ID                  int64
Customer_Name               str
Product                     str
Quantity                float64
Unit_Price              float64
Order_Date       datetime64[us]
Region                      str
dtype: object
```

In [20]:
```
df["Customer_Name"] = df["Customer_Name"].str.strip()
df["Product"] = df["Product"].str.strip()
df["Region"] = df["Region"].str.strip()
df[["Customer_Name", "Product", "Region"]]
```

Out[20]:

	Customer_Name	Product	Region
0	rahul sharma	laptop	hyderabad
1	PRIYA REDDY	MOBILE	KHAMMAM
2	Amit Kumar	headphones	warangal
3	sneha rao	TABLET	HYDERABAD
4	NaN	Laptop	khammam
5	KIRAN	mobile	Warangal
6	ananya reddy	Headphones	hyderabad
7	RAHUL SHARMA	laptop	KHAMMAM
8	arjun reddy	TABLET	warangal
9	meena devi	laptop	HYDERABAD
10		NaN	NaN
11	Vikram Rao	MOBILE	khammam
12	sai krishna	headphones	Warangal
13	Priya Reddy	tablet	hyderabad
14	naveen kumar	Laptop	KHAMMAM
15	divya sharma	mobile	warangal
17	Rohit Verma	LAPTOP	HYDERABAD

In [21]:

```
df["Customer_Name"] = df["Customer_Name"].str.title()
df["Product"] = df["Product"].str.title()
df["Region"] = df["Region"].str.title()

df[["Customer_Name", "Product", "Region"]]
```

Out[21]:

	Customer_Name	Product	Region
0	Rahul Sharma	Laptop	Hyderabad
1	Priya Reddy	Mobile	Khammam
2	Amit Kumar	Headphones	Warangal
3	Sneha Rao	Tablet	Hyderabad
4	NaN	Laptop	Khammam
5	Kiran	Mobile	Warangal
6	Ananya Reddy	Headphones	Hyderabad
7	Rahul Sharma	Laptop	Khammam
8	Arjun Reddy	Tablet	Warangal
9	Meena Devi	Laptop	Hyderabad
10		NaN	NaN
11	Vikram Rao	Mobile	Khammam
12	Sai Krishna	Headphones	Warangal
13	Priya Reddy	Tablet	Hyderabad
14	Naveen Kumar	Laptop	Khammam
15	Divya Sharma	Mobile	Warangal
17	Rohit Verma	Laptop	Hyderabad

In [22]:
```
text_columns = ["Customer_Name", "Product", "Region"]
df[text_columns] = df[text_columns].replace(
    r"^\s*$", np.nan, regex=True
)
df[text_columns].isnull().sum()
```

Out[22]:
```
Customer_Name    2
Product          1
Region           1
dtype: int64
```

In [23]:
```
df
```

Out[23]:

	Order_ID	Customer_Name	Product	Quantity	Unit_Price	Order_Date	Region
0	1001	Rahul Sharma	Laptop	2.0	55000.0	2026-05-01	Hyderabad
1	1002	Priya Reddy	Mobile	3.0	18000.0	2026-01-06	Khammam
2	1003	Amit Kumar	Headphones	5.0	1500.0	2026-07-01	Warangal
3	1004	Sneha Rao	Tablet	1.0	22000.0	2026-01-08	Hyderabad
4	1005	NaN	Laptop	4.0	55000.0	2026-09-01	Khammam
5	1006	Kiran	Mobile	NaN	18000.0	2026-01-10	Warangal
6	1007	Ananya Reddy	Headphones	2.0	1500.0	2026-11-01	Hyderabad
7	1008	Rahul Sharma	Laptop	3.0	NaN	2026-01-12	Khammam
8	1009	Arjun Reddy	Tablet	1.0	22000.0	NaT	Warangal
9	1010	Meena Devi	Laptop	2.0	55000.0	2026-01-14	Hyderabad
10	1011	NaN	NaN	6.0	18000.0	2026-01-15	NaN
11	1012	Vikram Rao	Mobile	4.0	NaN	2026-01-16	Khammam
12	1013	Sai Krishna	Headphones	NaN	1500.0	2026-01-17	Warangal
13	1014	Priya Reddy	Tablet	2.0	22000.0	2026-01-18	Hyderabad
14	1015	Naveen Kumar	Laptop	3.0	55000.0	2026-01-19	Khammam
15	1016	Divya Sharma	Mobile	5.0	1558000.0	2026-01-20	Warangal
17	1018	Rohit Verma	Laptop	2.0	99999.0	NaT	Hyderabad

In [24]:
```
df = df.dropna(subset=["Order_Date"])
print("Shape after removing invalid dates:", df.shape)
```

```
Shape after removing invalid dates: (15, 7)
```

In [25]:
```
df
```

Out[25]:

	Order_ID	Customer_Name	Product	Quantity	Unit_Price	Order_Date	Region
0	1001	Rahul Sharma	Laptop	2.0	55000.0	2026-05-01	Hyderabad
1	1002	Priya Reddy	Mobile	3.0	18000.0	2026-01-06	Khammam
2	1003	Amit Kumar	Headphones	5.0	1500.0	2026-07-01	Warangal
3	1004	Sneha Rao	Tablet	1.0	22000.0	2026-01-08	Hyderabad
4	1005	NaN	Laptop	4.0	55000.0	2026-09-01	Khammam
5	1006	Kiran	Mobile	NaN	18000.0	2026-01-10	Warangal
6	1007	Ananya Reddy	Headphones	2.0	1500.0	2026-11-01	Hyderabad
7	1008	Rahul Sharma	Laptop	3.0	NaN	2026-01-12	Khammam
9	1010	Meena Devi	Laptop	2.0	55000.0	2026-01-14	Hyderabad
10	1011	NaN	NaN	6.0	18000.0	2026-01-15	NaN
11	1012	Vikram Rao	Mobile	4.0	NaN	2026-01-16	Khammam
12	1013	Sai Krishna	Headphones	NaN	1500.0	2026-01-17	Warangal
13	1014	Priya Reddy	Tablet	2.0	22000.0	2026-01-18	Hyderabad
14	1015	Naveen Kumar	Laptop	3.0	55000.0	2026-01-19	Khammam
15	1016	Divya Sharma	Mobile	5.0	1558000.0	2026-01-20	Warangal

In [26]:
```
quantity_mean = df["Quantity"].mean()
price_mean = df["Unit_Price"].mean()
print("Quantity mean:", quantity_mean)
print("Unit price mean:", price_mean)
df["Quantity"] = df["Quantity"].fillna(quantity_mean)
df["Unit_Price"] = df["Unit_Price"].fillna(price_mean)
```

```
Quantity mean: 3.230769230769231
Unit price mean: 144653.84615384616
```

In [27]:
```
df["Customer_Name"] = df["Customer_Name"].fillna("Unknown")
df["Product"] = df["Product"].fillna("Unknown")
df["Region"] = df["Region"].fillna("Not Specified")
```

In [28]:
```
df.isnull().sum()
```

Out[28]:
```
Order_ID        0
Customer_Name   0
Product         0
Quantity        0
Unit_Price      0
Order_Date      0
Region          0
dtype: int64
```

In [29]:
```
df.dtypes
```

Out[29]:
```
Order_ID                  int64
Customer_Name               str
Product                     str
Quantity                float64
Unit_Price              float64
Order_Date       datetime64[us]
Region                      str
dtype: object
```

In [30]:
```
df.info()
```

```
<class 'pandas.DataFrame'>
Index: 15 entries, 0 to 15
Data columns (total 7 columns):
 #   Column         Non-Null Count  Dtype         
---  ------         --------------  -----         
 0   Order_ID       15 non-null     int64         
 1   Customer_Name  15 non-null     str           
 2   Product        15 non-null     str           
 3   Quantity       15 non-null     float64       
 4   Unit_Price     15 non-null     float64       
 5   Order_Date     15 non-null     datetime64[us]
 6   Region         15 non-null     str           
dtypes: datetime64[us](1), float64(2), int64(1), str(3)
memory usage: 960.0 bytes
```

In [31]:
```
df
```

Out[31]:

	Order_ID	Customer_Name	Product	Quantity	Unit_Price	Order_Date	Region
0	1001	Rahul Sharma	Laptop	2.000000	5.500000e+04	2026-05-01	Hyderabad
1	1002	Priya Reddy	Mobile	3.000000	1.800000e+04	2026-01-06	Khammam
2	1003	Amit Kumar	Headphones	5.000000	1.500000e+03	2026-07-01	Warangal
3	1004	Sneha Rao	Tablet	1.000000	2.200000e+04	2026-01-08	Hyderabad
4	1005	Unknown	Laptop	4.000000	5.500000e+04	2026-09-01	Khammam
5	1006	Kiran	Mobile	3.230769	1.800000e+04	2026-01-10	Warangal
6	1007	Ananya Reddy	Headphones	2.000000	1.500000e+03	2026-11-01	Hyderabad
7	1008	Rahul Sharma	Laptop	3.000000	1.446538e+05	2026-01-12	Khammam
9	1010	Meena Devi	Laptop	2.000000	5.500000e+04	2026-01-14	Hyderabad
10	1011	Unknown	Unknown	6.000000	1.800000e+04	2026-01-15	Not Specified
11	1012	Vikram Rao	Mobile	4.000000	1.446538e+05	2026-01-16	Khammam
12	1013	Sai Krishna	Headphones	3.230769	1.500000e+03	2026-01-17	Warangal
13	1014	Priya Reddy	Tablet	2.000000	2.200000e+04	2026-01-18	Hyderabad
14	1015	Naveen Kumar	Laptop	3.000000	5.500000e+04	2026-01-19	Khammam
15	1016	Divya Sharma	Mobile	5.000000	1.558000e+06	2026-01-20	Warangal

In [32]:
```
df = df.reset_index(drop=True)
df
```

Out[32]:

	Order_ID	Customer_Name	Product	Quantity	Unit_Price	Order_Date	Region
0	1001	Rahul Sharma	Laptop	2.000000	5.500000e+04	2026-05-01	Hyderabad
1	1002	Priya Reddy	Mobile	3.000000	1.800000e+04	2026-01-06	Khammam
2	1003	Amit Kumar	Headphones	5.000000	1.500000e+03	2026-07-01	Warangal
3	1004	Sneha Rao	Tablet	1.000000	2.200000e+04	2026-01-08	Hyderabad
4	1005	Unknown	Laptop	4.000000	5.500000e+04	2026-09-01	Khammam
5	1006	Kiran	Mobile	3.230769	1.800000e+04	2026-01-10	Warangal
6	1007	Ananya Reddy	Headphones	2.000000	1.500000e+03	2026-11-01	Hyderabad
7	1008	Rahul Sharma	Laptop	3.000000	1.446538e+05	2026-01-12	Khammam
8	1010	Meena Devi	Laptop	2.000000	5.500000e+04	2026-01-14	Hyderabad
9	1011	Unknown	Unknown	6.000000	1.800000e+04	2026-01-15	Not Specified
10	1012	Vikram Rao	Mobile	4.000000	1.446538e+05	2026-01-16	Khammam
11	1013	Sai Krishna	Headphones	3.230769	1.500000e+03	2026-01-17	Warangal
12	1014	Priya Reddy	Tablet	2.000000	2.200000e+04	2026-01-18	Hyderabad
13	1015	Naveen Kumar	Laptop	3.000000	5.500000e+04	2026-01-19	Khammam
14	1016	Divya Sharma	Mobile	5.000000	1.558000e+06	2026-01-20	Warangal

In [33]:
```
Q1 = df["Unit_Price"].quantile(0.25)
Q2 = df["Unit_Price"].quantile(0.50)
Q3 = df["Unit_Price"].quantile(0.75)
print("Q1:", Q1)
print("Q2 (Median):", Q2)
print("Q3:", Q3)
```

```
Q1: 18000.0
Q2 (Median): 22000.0
Q3: 55000.0
```

In [34]:
```
IQR = Q3 - Q1
print("IQR:", IQR)
```

```
IQR: 37000.0
```

In [35]:
```
lower_bound = Q1 - 1.5 * IQR
upper_bound = Q3 + 1.5 * IQR
print("Lower Bound:", lower_bound)
print("Upper Bound:", upper_bound)
```

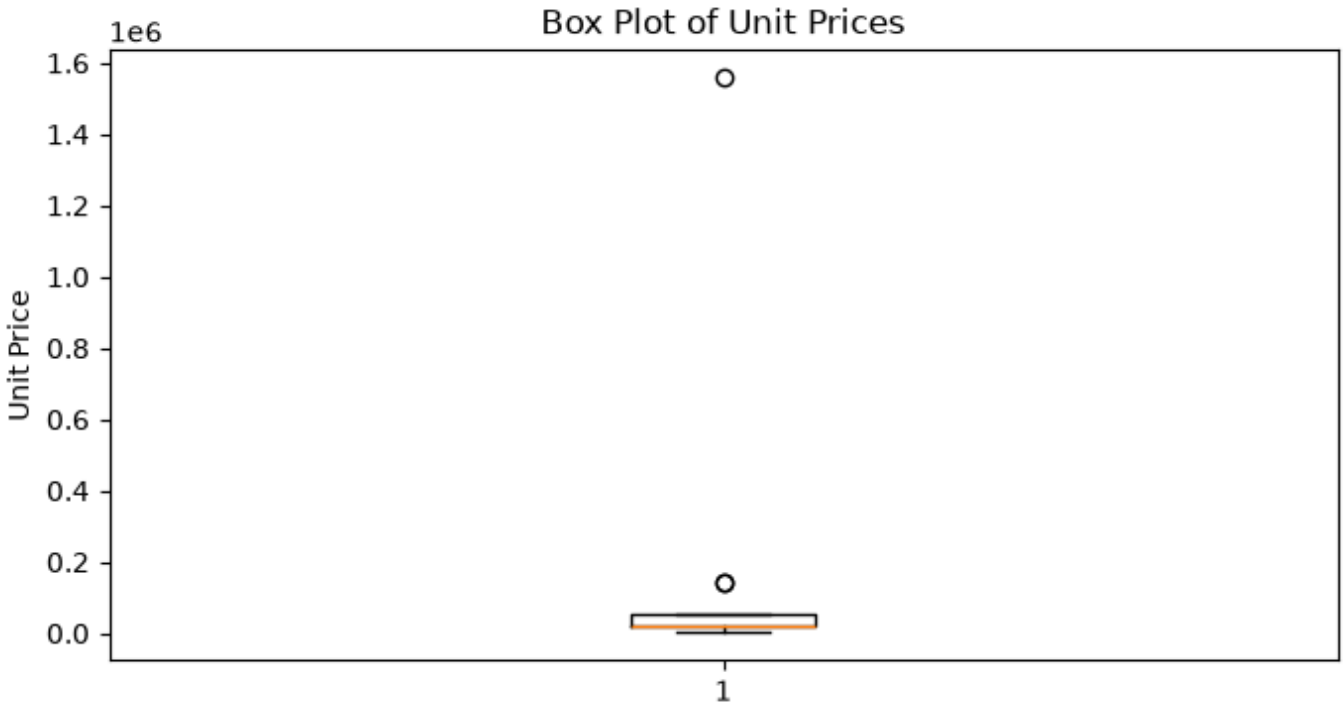
```
Lower Bound: -37500.0
Upper Bound: 110500.0
```

In [36]:
```
outliers = df[
    (df["Unit_Price"] < lower_bound) |
    (df["Unit_Price"] > upper_bound)
]
print(outliers[["Order_ID", "Product", "Unit_Price"]])
```

```
   Order_ID Product    Unit_Price
7      1008  Laptop  1.446538e+05
10     1012  Mobile  1.446538e+05
14     1016  Mobile  1.558000e+06
```

In [37]:
```
plt.figure(figsize=(8, 4))
plt.boxplot(df["Unit_Price"])
```

```
plt.title("Box Plot of Unit Prices")
plt.ylabel("Unit Price")
plt.show()
```



In [38]:
```
df
```

Out[38]:

	Order_ID	Customer_Name	Product	Quantity	Unit_Price	Order_Date	Region
0	1001	Rahul Sharma	Laptop	2.000000	5.500000e+04	2026-05-01	Hyderabad
1	1002	Priya Reddy	Mobile	3.000000	1.800000e+04	2026-01-06	Khammam
2	1003	Amit Kumar	Headphones	5.000000	1.500000e+03	2026-07-01	Warangal
3	1004	Sneha Rao	Tablet	1.000000	2.200000e+04	2026-01-08	Hyderabad
4	1005	Unknown	Laptop	4.000000	5.500000e+04	2026-09-01	Khammam
5	1006	Kiran	Mobile	3.230769	1.800000e+04	2026-01-10	Warangal
6	1007	Ananya Reddy	Headphones	2.000000	1.500000e+03	2026-11-01	Hyderabad
7	1008	Rahul Sharma	Laptop	3.000000	1.446538e+05	2026-01-12	Khammam
8	1010	Meena Devi	Laptop	2.000000	5.500000e+04	2026-01-14	Hyderabad
9	1011	Unknown	Unknown	6.000000	1.800000e+04	2026-01-15	Not Specified
10	1012	Vikram Rao	Mobile	4.000000	1.446538e+05	2026-01-16	Khammam
11	1013	Sai Krishna	Headphones	3.230769	1.500000e+03	2026-01-17	Warangal
12	1014	Priya Reddy	Tablet	2.000000	2.200000e+04	2026-01-18	Hyderabad
13	1015	Naveen Kumar	Laptop	3.000000	5.500000e+04	2026-01-19	Khammam
14	1016	Divya Sharma	Mobile	5.000000	1.558000e+06	2026-01-20	Warangal

In [39]:
```
df["Total_Sales"] = df["Quantity"] * df["Unit_Price"]
```

In [40]:
```
def categorize_sales(sales):
    if sales < 10000:
        return "Low Sales"
    elif sales < 50000:
        return "Medium Sales"
```

```
    else:
        return "High Sales"
```

In [41]:
```
df["Sales_Category"] = df["Total_Sales"].apply(categorize_sales)
```

In [42]:
```
df[["Order_ID", "Total_Sales", "Sales_Category"]]
```

Out[42]:

	Order_ID	Total_Sales	Sales_Category
0	1001	1.100000e+05	High Sales
1	1002	5.400000e+04	High Sales
2	1003	7.500000e+03	Low Sales
3	1004	2.200000e+04	Medium Sales
4	1005	2.200000e+05	High Sales
5	1006	5.815385e+04	High Sales
6	1007	3.000000e+03	Low Sales
7	1008	4.339615e+05	High Sales
8	1010	1.100000e+05	High Sales
9	1011	1.080000e+05	High Sales
10	1012	5.786154e+05	High Sales
11	1013	4.846154e+03	Low Sales
12	1014	4.400000e+04	Medium Sales
13	1015	1.650000e+05	High Sales
14	1016	7.790000e+06	High Sales

In [ ]:
```

```